

Rana Helal

AI & Machine Learning Engineer

ranaesmailhelal@gmail.com [+201272593734](https://www.linkedin.com/in/rana-helal) [Cairo, Egypt](#) [Portfolio](#) [LinkedIn](#) [github](#)

Summary	AI Systems Engineer with hands-on experience in building production-grade LLM systems, retrieval-augmented applications, and scalable AI pipelines. Experienced in designing backend-integrated AI solutions including fine-tuning, inference optimization, and graph-based recommendation systems.		
Experience	MasteryHub ITS		Cairo, Egypt
	Artificial Intelligence Engineer		Jan 2026 – Present
	- Developed and deployed LLM-powered AI systems with optimized inference - Built scalable FastAPI backend services and reliable AI pipelines		
	Artificial Intelligence Intern		Aug 2025 – Dec 2025
	Developed and integrated AI solutions across multiple projects based on business requirements, including NLP and computer vision tasks.		
	<u>Datalentech</u>		Cairo, Egypt
	AI Developer Intern		Mar 2026 – Apr 2026
Built an embedding-based recommendation system with scalable FastAPI architecture and dynamic skill taxonomy design for personalized AI/ML learning paths			
	National Telecommunication Institute (NTI)		Cairo, Egypt
	Computer Vision Intern – AI Summer Internship Program		Jul 2025 – Aug 2025
	Implemented CNNs, ResNet, and DenseNet for image classification.		
	NTI - Huawei Egyptian Talents Academy		Cairo, Egypt
AI Trainee		Apr 2025 – Jul 2025	
Trained in Python, data analysis, machine learning, and deep learning.			
Education	Cairo University	Bachelor's Degree. • 3.66 (Ranked 3rd in AI Department)	
	Computer Science and Artificial Intelligence		Oct 2022 – Jul 2026
	Relevant Coursework:	Data Structures, Software Development, Algorithms, Machine Learning	
Projects	SpeakOut AI Content Moderation System		Dec 2025 – Present
	- Built and fine-tuned LLMs using SFTTrainer with LoRA-based adaptation, through 4-bit quantization and mixed precision (FP16/BF16) - Developed a lightweight local-first moderation model optimized for low-latency inference, integrated with the frontend for real-time user feedback using ONNX (INT8 quantization) - Engineered GPU lifecycle management and cron-based scheduling system with queue-driven activation logic to optimize resource utilization and reduce idle compute time - Implemented high-throughput batch processing pipeline handling up to 100 posts per batch with optimized inference efficiency (~0.13–0.15s per post)		
	Graph-Based Learning Recommendation System		Mar 2026 – April 2026
	- Built a graph-based learning path system using DFS and topological sorting to generate structured skill sequences based on prerequisite relationships - Designed an embedding-based ranking engine combining cosine similarity, skill weights, and content quality scoring for personalized recommendations - Implemented a dynamic content pipeline with external fetching, database expansion, and filtering to maintain up-to-date learning resources		
	<u>LLM-based Agentic RAG System for Automated Presentation Generation</u>		Mar 2026
	- Built an AI presentation agent that generates structured slide content and narration using LLM-based multimodal pipelines - Designed and implemented an agentic RAG system using LangChain, integrating retrieval, tool usage, and session-aware memory for real-time content generation - Developed a Text-to-Speech service using Gemini TTS, supporting single and multi-speaker voice synthesis for dynamic presentation delivery		
	<u>AI Agent for Diabetes Diagnosis</u>		Sep 2025
Fine-tuned LLaMA and BioGPT on a custom medical dataset, deployed via FastAPI and Gradio			

Skills	Machine Learning & AI:	Programming & Development	Cloud & Databases:
	Deep Learning, LLM Fine-tuning, NLP & Text Classification, Computer Vision, Model Evaluation	Python, C++, Dart, TypeScript, NestJS, FastAPI, Flutter, Firebase	AWS S3, AWS SageMaker, Vector Databases, Microsoft SQL Server